

A Mathematical Proof of Mode Collapse as an Inherent Failure Mode in the Generative Adversarial Network Framework

Abstract

This report provides a rigorous mathematical proof of the existence of mode collapse in Generative Adversarial Networks (GANs). We begin by formalizing the GAN as a minimax game and demonstrate that its objective is equivalent to minimizing the Jensen-Shannon (JS) divergence between the real and generated data distributions. We then invoke the manifold hypothesis to argue that in high-dimensional spaces, the supports of these distributions are almost certainly disjoint. Under this condition, we prove that the JS divergence becomes a constant, causing the gradient of the generator's loss function to vanish as the discriminator approaches optimality. This creates a pathological optimization landscape where the generator is incentivized to exploit narrow regions of non-zero gradient, leading to a stable equilibrium state characterized by a low-diversity output distribution. We formally define this state as mode collapse and contrast the underlying mechanics with the Wasserstein distance, whose continuity under disjoint supports resolves this failure mode, thereby corroborating our proof that mode collapse is an inherent flaw in the original GAN formulation.

1. The Theoretical Framework of Generative Adversarial Networks

The introduction of Generative Adversarial Networks (GANs) by Goodfellow et al. in 2014 marked a paradigm shift in generative modeling.¹ Rather than relying on explicit maximum likelihood estimation, which often involves intractable probabilistic computations, the GAN

framework learns to generate data through a competitive, game-theoretic process.² This section establishes the formal mathematical foundations of this framework, which are essential for the subsequent proof of its inherent instabilities.

1.1. The GAN as a Two-Player Minimax Game

At its core, a GAN is a system composed of two neural networks, a generator and a discriminator, locked in a zero-sum game.¹

The Generator, denoted as G , is a differentiable function parameterized by a vector θ . It is designed to map points from a simple, well-defined latent space to the more complex data space. Formally, we define this mapping as:

$$G: \mathcal{Z} \rightarrow \mathcal{X}$$

where $\mathcal{Z}$ is the latent space (typically a low-dimensional Euclidean space) and $\mathcal{X}$ is the data space (e.g., the space of images). The generator takes as input a latent vector z sampled from a prior distribution $p_z(z)$ (e.g., a standard multivariate Gaussian or uniform distribution) and outputs a sample $x = G(z)$ that is intended to resemble real data.⁵

The **Discriminator**, denoted as D , is a second differentiable function, parameterized by a vector w . Its role is to act as a binary classifier. It takes a sample x from the data space $\mathcal{X}$ and outputs a scalar value representing the probability that x is a sample from the true data distribution rather than one produced by the generator.² The mapping is formally defined as:

$$D: \mathcal{X} \rightarrow [0, 1]$$

An output $D(x)$ close to 1 signifies that the discriminator believes x is real, while an output close to 0 indicates it believes x is fake (i.e., generated).⁶

The training process is structured as a two-player minimax game defined by a single value function, $V(G, D)$.¹ This function is derived from the cross-entropy loss for a binary classifier:

$$V(G, D) = \mathbb{E}_{x \sim p_{\text{data}}(x)} \log D(x) + \mathbb{E}_{z \sim p_z(z)} \log (1 - D(G(z)))$$

Here, $p_{\text{data}}(x)$ is the true data distribution and $p_z(z)$ is the prior distribution over the latent variables. The discriminator, D , seeks to maximize this value function. The first term, $\mathbb{E}_{x \sim p_{\text{data}}(x)} \log D(x)$, is maximized when $D(x)$ is close to 1 for real samples. The second term, $\mathbb{E}_{z \sim p_z(z)} \log (1 - D(G(z)))$, is maximized when $D(G(z))$ is close to 0 for generated samples. Concurrently, the generator, G , seeks to minimize the value function by producing samples $G(z)$ that fool the discriminator into outputting a high probability, i.e., making $D(G(z))$ close to 1, which in turn minimizes $\log(1 - D(G(z)))$.⁷

This adversarial dynamic is captured by the minimax objective ³:

$$\min_G \max_D V(G, D)$$

The theoretical equilibrium of this game is a Nash equilibrium where the generator produces samples indistinguishable from real data, and the discriminator is maximally confused, outputting a probability of 1/2 for all samples.³

1.2. The Role of Probability Distributions

The ultimate goal of the GAN framework is to learn the underlying probability distribution of the training data. Let $p_{data}(x)$ be the true, unknown probability density function of the real data. The generator G implicitly defines a probability distribution over the data space, which we denote as $p_G(x)$. This distribution is formed by taking an infinite number of samples $z \sim p_Z(z)$ and transforming them through the generator, $x = G(z)$. The objective of the generator is to adjust its parameters θ such that the generated distribution p_G becomes identical to the true data distribution p_{data} .¹

A critical characteristic of the GAN framework is that it is an *implicit* generative model.¹ Unlike models that explicitly define and maximize a likelihood function

$p_G(x; \theta)$, a GAN does not provide a way to evaluate the density $p_G(x)$ for a given point x . It only provides a mechanism to *sample* from this distribution by passing a latent vector z through the generator network.² This distinction is of paramount importance. Because the density functions are intractable, the optimization of the value function

$V(G, D)$ must be performed using sample-based approximations (Monte Carlo estimates) via mini-batches of real and generated data. This reliance on comparing finite sets of samples, rather than analytically comparing the full distributions, makes the entire learning process highly sensitive to the geometric arrangement of these sample sets in the high-dimensional data space. This sensitivity is a foundational element in the emergence of mode collapse, a connection that is not immediately apparent from the value function's formulation alone.

2. The Optimal Discriminator and the Jensen-Shannon Divergence Objective

To understand what the generator is truly optimizing, we must first analyze the inner loop of

the minimax game: the maximization of $V(G,D)$ with respect to the discriminator D for any fixed generator G . This analysis reveals that the complex game-theoretic objective simplifies to a well-known statistical divergence minimization problem, thereby exposing the underlying metric that governs the generator's learning process.

2.1. Derivation of the Optimal Discriminator $D^*(x)$

For any given generator G , which defines a fixed distribution p_g , the optimal discriminator DG^* is the function that maximizes $V(G,D)$. We can express the value function as an integral over the data space X by expanding the expectations⁸:

$$V(G,D) = \int p_{data}(x) \log(D(x)) dx + \int p_g(z) \log(1-D(G(z))) dz$$

Using the change of variables formula, the second integral can be rewritten in terms of an expectation over the generator's distribution p_g .⁸ This allows us to combine both terms under a single integral over x :

$$V(G, D) = \int p_{data}(x) \log(D(x)) dx + \int p_g(x) \log(1-D(x)) dx$$

To find the discriminator that maximizes this value, we can perform a point-wise maximization of the integrand for each x . Let us consider the function $f(y) = a \log(y) + b \log(1-y)$, where $y = D(x)$, $a = p_{data}(x)$, and $b = p_g(x)$. This function is a standard form related to the binary cross-entropy. To find its maximum, we take the derivative with respect to y and set it to zero⁸:

$$\frac{df}{dy} = \frac{a}{y} - \frac{b}{1-y} = 0$$

Solving for y yields:

$$a(1-y) = by \Rightarrow a - ay = by \Rightarrow a = y(a+b) \Rightarrow y = \frac{a}{a+b}$$

Substituting back $p_{data}(x)$ for a and $p_g(x)$ for b , we obtain the form of the optimal discriminator 1:

$$D^*(x) = \frac{p_{data}(x)}{p_{data}(x) + p_g(x)}$$

This elegant result demonstrates that the optimal discriminator for a fixed generator perfectly estimates the probability that a given sample x belongs to the real data distribution. If $p_{data}(x) \gg p_g(x)$, $D^*(x) \rightarrow 1$. If $p_g(x) \gg p_{data}(x)$, $D^*(x) \rightarrow 0$. If $p_{data}(x) = p_g(x)$, then $D^*(x) = 1/2$, signifying maximal uncertainty.⁸

2.2. Transforming the Generator's Objective into Divergence

Minimization

With the optimal discriminator $D^*(x)$ identified, we can now substitute it back into the value function to determine the objective that the generator aims to minimize. This objective is $C(G) = \max_D V(G, D) = V(G, D^*)$.¹⁰

$$C(G) = \mathbb{E}_{x \sim p_{\text{data}}} [\log p_{\text{data}}(x) + p_g(x) p_{\text{data}}(x)] + \mathbb{E}_{x \sim p_g} [\log p_{\text{data}}(x) + p_g(x) p_g(x)]$$

To see the connection to standard information-theoretic divergences, we first define the Kullback-Leibler (KL) divergence and the Jensen-Shannon (JS) divergence. The KL divergence between two distributions P and Q is:

$$KL(P \parallel Q) = \int P(x) \log Q(x) P(x) dx$$

The Jensen-Shannon divergence is a symmetrized and smoothed version of the KL divergence¹²:

$$JSD(P \parallel Q) = \frac{1}{2} KL(P \parallel \frac{P+Q}{2}) + \frac{1}{2} KL(Q \parallel \frac{P+Q}{2})$$

Let's now manipulate the expression for $C(G)$. We can add and subtract $\log(2)$ within the logarithms:

$$C(G) = \mathbb{E}_{x \sim p_{\text{data}}} [\log \frac{p_{\text{data}}(x) + p_g(x)}{2} p_{\text{data}}(x) - \log(2)] + \mathbb{E}_{x \sim p_g} [\log \frac{p_{\text{data}}(x) + p_g(x)}{2} p_g(x) - \log(2)]$$

Recognizing the KL divergence terms, we have:

$$C(G) = KL\left(p_{\text{data}} \parallel \frac{p_{\text{data}} + p_g}{2}\right) + KL\left(p_g \parallel \frac{p_{\text{data}} + p_g}{2}\right) - 2\log(2)$$

This simplifies to the final, pivotal result 1:

$$C(G) = 2 \cdot JSD(p_{\text{data}} \parallel p_g) - 2\log(2)$$

This derivation proves that the minimax objective of the GAN framework is equivalent to minimizing the Jensen-Shannon divergence between the true data distribution and the generated distribution, up to a scaling factor and a constant. The global minimum is achieved when $p_g = p_{\text{data}}$, at which point $JSD(p_{\text{data}} \parallel p_g) = 0$ and $C(G) = -2\log(2)$.⁸

It is crucial to recognize that this equivalence hinges on the assumption of an optimal discriminator. This is a theoretical idealization that abstracts away the practical reality of simultaneous, iterative training.¹⁵ In practice, the discriminator is never fully optimal at any given step; it is a co-evolving adversary. The generator's gradients are therefore calculated with respect to a constantly changing, imperfect discriminator. However, the

asymptotic landscape that the generator attempts to navigate is defined by the Jensen-Shannon divergence. Consequently, any fundamental pathologies in the JSD as a loss function will manifest as severe instabilities in the practical training algorithm. The proof of equivalence to JSD minimization thus provides the crucial lens through which we can analyze and predict the failures of the GAN framework.¹⁵

3. The Manifold Hypothesis and the Problem of Disjoint Supports

The theoretical elegance of the JSD minimization objective belies a profound practical problem that arises from the geometry of high-dimensional data. The mathematical properties of the JSD are well-behaved when the distributions being compared have overlapping support. However, in the context of GANs generating complex data like images, this assumption of overlap is almost certainly violated.

3.1. The Manifold Hypothesis

High-dimensional data distributions are rarely, if ever, uniform. Natural data, such as images, audio, or text, possess a high degree of internal structure and correlation. This structure constrains the data to lie on or near a low-dimensional manifold embedded within the high-dimensional ambient space.¹⁸ For instance, the space of all possible

64×64 pixel images is vast (2564096-dimensional for 8-bit grayscale), but the subset of these images that look like a human face is an infinitesimally small, highly structured manifold within that space.

The generator network, by its very construction, reinforces this property. The generator function G maps a low-dimensional latent space Z (e.g., with dimension $d_z=100$) to a high-dimensional data space X (e.g., with dimension $d_x=3072$ for a 32×32×3 color image). As a continuous mapping from a lower-dimensional space to a higher-dimensional one, the output of the generator, and thus the support of the distribution p_g , is necessarily confined to a manifold whose intrinsic dimension is at most d_z .⁹

3.2. Disjoint Supports as a High-Probability Event

Given that both the true data distribution p_{data} and the generated distribution p_g are supported on low-dimensional manifolds, the probability of these two manifolds intersecting substantially in a high-dimensional space is negligible.¹⁸ Two lines randomly placed in a 3D space are unlikely to intersect; two surfaces are even less likely to have a significant overlap. As the dimension of the ambient space grows, the probability of intersection plummets.

Formally, this means that the supports of p_{data} and p_g are almost certainly disjoint. The support of a distribution is the set of points where its density is non-zero. Disjoint supports imply that for almost every point $x \in X$, if $p_{\text{data}}(x) > 0$, then $p_g(x) = 0$, and conversely, if $p_g(x) > 0$, then $p_{\text{data}}(x) = 0$. This condition is particularly acute in the early stages of training, when the generator's output is essentially random and has no reason to align with the true data manifold.¹⁶

This geometric reality exposes a deep and critical flaw in the original GAN theory. The very derivation of the optimal discriminator D^* in Section 2.1 relies on the calculus of variations, which assumes that the integrand is continuously differentiable. However, when the data dimension d_x is greater than the latent dimension d_z , the probability density function $p_g(x)$ is degenerate—it contains Dirac delta functions and is not a well-behaved, continuous function over the ambient space X . This degeneracy means the integrand in the value function is not continuously differentiable, and the derivation of D^* via simple calculus is mathematically invalid in the exact setting where GANs are most commonly applied.⁹ This fundamental inconsistency between the mathematical assumptions and the practical application domain is the root cause of the training instabilities. While the practical algorithm, using powerful neural network function approximators, can smooth over these theoretical discontinuities to some extent, the underlying mathematical pathology inevitably surfaces as catastrophic training failures, such as the vanishing gradient problem.

4. Mathematical Proof of Gradient Vanishing with Jensen-Shannon Divergence

This section presents the central argument of the report: a formal proof that the combination of the JSD objective and the geometric reality of disjoint supports leads to a vanishing gradient for the generator. This demonstrates that the learning process is destined to fail under the very conditions it is designed to operate in.

4.1. The Constant Value of JSD for Disjoint Distributions

As established, when the supports of p_{data} and p_g are disjoint, for any point x , at most one of the densities can be non-zero. Let's analyze the value of the JSD under this condition. The JSD is defined as:

$$JSD(p_{data} // p_g) = \frac{1}{2} KL(p_{data} // M) + \frac{1}{2} KL(p_g // M)$$

where $M = p_{data} + p_g$.

Let's evaluate the first term, $KL(p_{data} // M)$:

$$KL(p_{data} // M) = \int p_{data}(x) \log M(x) p_{data}(x) dx$$

The integral is taken over the support of p_{data} . Within this support, $p_{data}(x) > 0$ and, due to disjointness, $p_g(x) = 0$. Therefore, inside the integral, $M(x) = p_{data}(x) + 0 = p_{data}(x)$.

Substituting this into the integrand gives:

$$p_{data}(x) \log \frac{p_{data}(x)}{p_{data}(x) + 0} = p_{data}(x) \log \frac{p_{data}(x)}{p_{data}(x)} = p_{data}(x) \log(1) = 0$$

The integral then becomes:

$$KL(p_{data} // M) = \int p_{data}(x) \log(2) dx = \log(2) \int p_{data}(x) dx = \log(2)$$

since $\int p_{data}(x) dx = 1$.

By a symmetrical argument for the second term, $KL(p_g // M)$, the integral is taken over the support of p_g , where $p_g(x) > 0$ and $p_{data}(x) = 0$. This also yields $\log(2)$.

Therefore, the Jensen-Shannon divergence between two distributions with disjoint supports is ¹³:

$$JSD(p_{data} // p_g) = \frac{1}{2} \log(2) + \frac{1}{2} \log(2) = \log(2)$$

This is a constant value. Since the generator's objective is $C(G) = 2 \cdot JSD(p_{data} // p_g) - 2 \log(2)$, under the condition of disjoint supports, the objective function becomes:

$$C(G) = 2 \cdot \log(2) - 2 \log(2) = 0$$

The generator's loss function becomes a constant, which means its gradient with respect to the generator's parameters θ must be zero. The loss landscape is perfectly flat, providing no signal for learning.

4.2. The Perfect Discriminator and Saturated Loss

The condition of disjoint supports has a direct consequence for the discriminator. It implies the existence of a perfect discriminator that can separate the two distributions with 100% accuracy. A neural network with sufficient capacity can learn to approximate this perfect classifier, outputting $D(x) \approx 1$ for all real samples and $D(x) \approx 0$ for all generated samples.¹⁸ This state, where the discriminator's loss is minimized, is often referred to as a "saturated" state, particularly because the sigmoid activation function at the final layer of the discriminator is pushed to its flat regions (0 and 1).¹⁹

4.3. Derivation of the Vanishing Gradient

Let's now formally derive the vanishing gradient for the generator. The generator's loss function, in its most common implementation, aims to maximize the discriminator's error on fake samples. This is typically written as¹:

$$L_G = \mathbb{E}_{z \sim p_z(z)} [D(G(z))]$$

The generator updates its parameters θ by descending the gradient of this loss. The gradient $\nabla_{\theta} L_G$ is calculated using the chain rule:

$$\nabla_{\theta} L_G = \mathbb{E}_{z \sim p_z(z)} [\nabla_{\theta} D(G(z))]$$

As the discriminator D approaches optimality (perfection), its output for any generated sample $G(z)$ approaches 0. Let's analyze the terms in the gradient expression in this limit:

1. The term $1 - D(G(z))$ approaches $1 - 0 = 1$.
2. The term $\nabla_{\theta} G(z)$ is the Jacobian of the generator network, which is generally non-zero.
3. The critical term is $\nabla D(G(z))$, which is the gradient of the discriminator's output with respect to its input, evaluated at the location of a generated sample.

The discriminator typically uses a sigmoid activation function $\sigma(a) = \frac{1}{1 + e^{-a}}$ to produce its final probability output. Its derivative is $\sigma'(a) = \sigma(a)(1 - \sigma(a))$. As $D(x) = \sigma(a) \rightarrow 0$, the pre-activation value a must go to $-\infty$. In this region, the derivative $\sigma'(a)$ approaches 0. This means the gradient of the discriminator's output becomes vanishingly small in the regions of the data space where it confidently classifies samples as fake.¹⁹ Consequently, the term

$\nabla D(G(z))$ approaches zero.

Therefore, the entire gradient expression for the generator collapses to zero ¹⁵:

$$D \rightarrow D^* \lim \nabla_{\theta} LG = 0$$

This is the mathematical proof of the vanishing gradient problem. A common heuristic to combat this is the "non-saturating" loss function, $LG = -\mathbb{E}_z \log p_D(z)$.²³ This formulation provides stronger gradients early in training when $D(G(z))$ is small. However, it does not resolve the fundamental issue that the underlying JSD landscape is flat. It merely changes the dynamics on that landscape, often leading to unstable oscillations and mode hopping rather than stable convergence to the true data distribution.²³

This vanishing gradient is not merely a numerical inconvenience; it represents a fundamental breakdown in the flow of information within the GAN framework. An optimal discriminator, by perfectly separating real and fake data, essentially provides a binary signal to the generator: "your samples are bad." It fails, however, to provide any information on *how* to improve them. The gradient is the mathematical embodiment of this "how-to" information. When the gradient vanishes, the flow of useful information from the discriminator to the generator ceases, and learning halts.

5. The Dynamics of Optimization and the Emergence of Mode Collapse

The mathematical proof of vanishing gradients establishes the existence of a pathological optimization landscape. This section connects this theoretical result to the observable, practical failure mode known as mode collapse, showing how the dynamics of gradient-based optimization on this landscape naturally lead to a low-diversity generative model.

5.1. The Pathological Optimization Landscape

The vanishing gradient proof implies that for a well-trained discriminator, the generator's loss surface is almost entirely flat. There are no informative gradients to guide the generator's parameters θ towards a state where p_G would cover the full diversity of p_D . This is not a landscape characterized by numerous poor local minima; rather, it is a landscape largely devoid of any useful features for optimization.²¹ The generator is effectively lost in a vast, featureless desert.

5.2. The Path of Least Resistance and Optimization Traps

In practice, the discriminator is not perfectly optimal at all times. During the iterative training process, there may be moments or regions in the data space where the discriminator is momentarily weak or uncertain. Suppose the generator, by chance, produces a sample $x' = G(z')$ that falls into one of these regions, where $D(x')$ is not yet saturated at 0.

This single sample, or a small cluster of similar samples, becomes one of the few places in the entire data space that provides a strong, non-zero gradient back to the generator. The optimization algorithm, such as stochastic gradient descent, is designed to follow the steepest descent. It will therefore aggressively update the generator's parameters θ to produce more samples similar to x' , as this is the most efficient—and perhaps only—way to decrease its loss and "fool" the current discriminator.²¹ The generator has found a "path of least resistance" and will exploit it relentlessly.

5.3. Formalizing Mode Collapse as a Stable Equilibrium

This dynamic leads directly to the phenomenon of mode collapse. We can formally define mode collapse as a state where the generator function G_θ maps a large-volume subset of the latent space Z into an arbitrarily small, low-variance region in the data space X . This small region typically corresponds to one, or a very limited number, of the modes (i.e., distinct clusters or patterns) present in the true data distribution p_{data} .²⁷ The generator learns to produce only a few types of outputs, ignoring the full diversity of the training data.³¹

This collapsed state represents a stable, albeit undesirable, equilibrium in the GAN's training dynamics. The process unfolds as follows:

1. The generator discovers a single mode (e.g., a specific digit in MNIST) that is easy to produce and momentarily fools the discriminator.
2. The generator's parameters are updated to concentrate the output distribution p_g onto this single mode.
3. The discriminator, in subsequent training steps, quickly learns to identify this specific, repetitive mode as "fake." Its decision boundary sharpens around this mode.
4. The discriminator once again becomes near-optimal with respect to the collapsed generator distribution.
5. This leads back to the vanishing gradient problem. The generator is now trapped. It receives no gradient signal to encourage it to move its parameters and explore other

modes of the data distribution.²¹

This cycle can sometimes manifest as mode hopping, where the generator collapses to one mode, gets penalized, then jumps to another mode it can easily exploit, and so on. This behavior, sometimes called the "Helvetica scenario," results in a generator that is always mode-collapsed, even if the specific mode changes over time, never learning to capture the full data distribution.²⁷

This process can be viewed as a form of catastrophic forgetting applied to the data distribution itself.²¹ As the generator's optimization process focuses intensely on the gradients provided by the current state of the discriminator (which may be focused on penalizing a specific mode), it overwrites the network parameters that may have encoded information about other, previously explored modes. The knowledge of the data distribution's diversity is actively destroyed by the optimization dynamic, which is driven by the local, myopic information provided by the discriminator's gradient.

6. Contrasting with the Wasserstein Distance: A Theoretical Corroboration

The proof that mode collapse is an inherent failure of the original GAN formulation can be powerfully corroborated by examining an alternative framework that resolves the underlying mathematical pathology. The introduction of the Wasserstein GAN (WGAN) provided precisely this contrast, demonstrating that by replacing the Jensen-Shannon divergence with a different distance metric—the Wasserstein distance—the vanishing gradient problem could be overcome.²⁰ This serves as a "control experiment," confirming that the flaw lies not in the adversarial concept itself, but in the specific choice of divergence metric.

6.1. The Earth Mover's (Wasserstein-1) Distance

The Wasserstein-1 distance, also known as the Earth Mover's Distance, provides a different way to measure the dissimilarity between two probability distributions. Intuitively, it measures the minimum "cost" required to transform one distribution into the other, where cost is defined as the amount of "mass" moved multiplied by the distance it is moved.¹³ Formally, it is defined as:

$$W(p_{data}, p_g) = \inf_{\gamma \in \Pi(p_{data}, p_g)} \int \int \|x - y\| d\gamma(x, y)$$

where $\Pi(p_{data}, p_g)$ is the set of all joint distributions $\gamma(x, y)$ whose marginal distributions are p_{data} and p_g , respectively.

Directly computing this infimum is intractable. However, the Kantorovich-Rubinstein duality provides an alternative, more practical formulation²⁰:

$$W(p_{data}, p_g) = \sup_{f: \mathbb{R}^d \rightarrow \mathbb{R}, L \leq 1} (E_{x \sim p_{data}}[f(x)] - E_{x \sim p_g}[f(x)])$$

Here, the supremum is taken over all 1-Lipschitz functions $f: X \rightarrow \mathbb{R}$. A function is 1-Lipschitz if $|f(x_1) - f(x_2)| \leq \|x_1 - x_2\|$ for all x_1, x_2 . In the WGAN framework, the discriminator is repurposed as a "critic" that is trained to approximate this function f .²⁰

6.2. Properties of Wasserstein Distance Under Disjoint Supports

The Wasserstein distance has fundamentally different mathematical properties from the JSD, particularly under the condition of disjoint supports.

- **Continuity:** Unlike the JSD, which is discontinuous (jumping from $\log(2)$ to 0 when distributions overlap), the Wasserstein distance is continuous. If two distributions supported on low-dimensional manifolds are moved closer together, the cost of transport smoothly decreases to zero. This provides a continuous loss surface that is amenable to gradient-based optimization.¹³
- **Meaningful Gradients:** Because the critic f is constrained to be 1-Lipschitz, it cannot saturate in the same way a standard discriminator does. The Lipschitz constraint prevents the function's gradient from becoming arbitrarily steep, forcing it to behave at most linearly. This ensures that the critic provides meaningful, non-zero gradients almost everywhere, even when it is trained to optimality. An optimal critic provides a better, more reliable gradient to the generator, directly resolving the vanishing gradient problem.²⁰ Training the critic to optimality is not only possible but beneficial in WGANs, a direct reversal of the situation in the original GAN framework.¹³

The choice of divergence metric can be seen as implicitly defining the geometry, or topology, of the space of probability distributions. The JSD induces a strong topology where distributions with disjoint supports are treated as being maximally and equally far apart, creating a landscape of disconnected "islands" with cliffs between them. Gradient descent cannot navigate such a space. The Wasserstein distance induces a weaker, smoother topology that respects the underlying geometry of the data space itself. It effectively builds bridges between the islands, creating a connected landscape with sloped paths that the generator's optimization process can successfully traverse.

6.3. A Comparative Analysis of Divergence Metrics

The following table summarizes the critical differences between the Jensen-Shannon divergence and the Wasserstein distance in the context of GAN training, highlighting why the latter provides a more stable foundation.

Property	Jensen-Shannon Divergence (Original GAN)	Wasserstein-1 Distance (WGAN)
Value with Disjoint Supports	Constant ($\log 2$)	Meaningful, non-zero value proportional to distance
Continuity of Loss w.r.t. θ	Discontinuous	Continuous (if G is continuous)
Generator Gradient	Vanishes as D becomes optimal	Becomes more reliable as critic becomes optimal
Discriminator/Critic Output	Probability (Sigmoid activation)	Unbounded Score (No activation)
Discriminator/Critic Saturation	Prone to saturation, causing flat gradients	No saturation due to Lipschitz constraint
Induced Topology	Strong, leads to disconnected spaces	Weaker, respects underlying metric space geometry
Primary Failure Mode	Vanishing Gradients & Mode Collapse	Can still suffer mode collapse (less severe), requires careful constraint (e.g., Gradient Penalty)

This comparative analysis solidifies the conclusion that the pathologies of the original GAN framework are direct consequences of the mathematical properties of the Jensen-Shannon divergence.

7. Concluding Analysis on the Inherent Instability of the GAN Framework

This report has constructed a step-by-step mathematical proof demonstrating that mode collapse is not an incidental bug or an artifact of implementation, but rather an inherent and predictable failure mode of the original Generative Adversarial Network formulation.

7.1. Summary of the Proof

The logical chain of the proof is as follows:

1. The GAN is formalized as a minimax game between a generator and a discriminator. We proved that, under the assumption of an optimal discriminator, this game is equivalent to the generator minimizing the **Jensen-Shannon (JS) divergence** between the true data distribution, p_{data} , and the generated distribution, p_g .
2. We invoked the **manifold hypothesis**, which posits that high-dimensional data like images reside on low-dimensional manifolds. This geometric reality implies that the supports of p_{data} and p_g are almost certainly **disjoint**, especially early in training.
3. We then proved that for distributions with disjoint supports, the JS divergence takes on a **constant value of $\log(2)$** . This renders the generator's objective function a constant, meaning its loss landscape is perfectly flat.
4. This flat landscape leads directly to the **vanishing gradient problem**. As the discriminator learns to perfectly separate the disjoint distributions, its output saturates, and the gradient signal it passes back to the generator becomes zero, halting the learning process.
5. Finally, we showed how this pathological optimization landscape leads to **mode collapse**. Gradient-based optimization will aggressively exploit any small, transient imperfection in the discriminator, causing the generator to concentrate all its probability mass on a single, easy-to-produce output, resulting in a stable but useless equilibrium.

The contrast with the Wasserstein GAN framework serves as a powerful corroboration of this conclusion. By replacing the JSD with the Wasserstein distance—a metric that is continuous for disjoint supports and provides non-vanishing gradients—the primary cause of training instability is removed.

7.2. Final Verdict

The conclusion is inescapable: mode collapse is a mathematically demonstrable and inherent consequence of applying the original GAN objective, which relies on the Jensen-Shannon divergence, to data distributions on low-dimensional manifolds. The theoretical elegance of the framework, which promises a model that can learn any data distribution, breaks down when confronted with the geometric realities of the high-dimensional spaces in which it is most often applied.

The extensive body of research on GAN stabilization techniques—from architectural changes to alternative loss functions and regularization schemes—can be understood as a continuous and varied effort to repair this fundamental flaw in the original formulation.¹⁷ While these methods have led to remarkable empirical successes, the analysis presented herein demonstrates that the instability they seek to correct is not an anomaly but a direct consequence of the foundational mathematics of the first generative adversarial networks.

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